

Artificial Intelligence & Machine Learning Architecture Specification

Mathematical Formulations, Training Regimes, Feature Engineering & Hyperparameter Optimization

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Machine Learning System Architecture

RentAI leverages machine learning across three distinct decision-support domains: customer attrition risk mitigation, multi-paradigm item recommendation, and localized conversational customer service.

2. Pipeline A: LightGBM Customer Churn Mitigation

2.1 Mathematical Formulation

LightGBM minimizes a binary cross-entropy loss function over an ensemble of M decision trees:

$$L(y, F(x)) = - \sum [y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i)]$$

LightGBM accelerates split finding via **Gradient-Based One-Side Sampling (GOSS)**, retaining instances with large gradients $|g_i|$ while sampling instances with small gradients, and **Exclusive Feature Bundling (EFB)** to compress sparse categorical vectors.

2.2 Engineered Telemetry Features (15 Attributes)

Feature Name	Formula / Origin	Significance
inactivity_ratio	Days inactive / Total tenure days	Primary indicator of waning interest
delinquency_rate	Overdue days / Total billings	Direct signal of financial or service distress
early_return_rate	Premature terminations / Total leases	Dissatisfaction with appliance performance
ticket_frequency	Support complaints / Active tenure months	Customer support friction indicator
cart_friction	Abandoned checkouts / Total sessions	Pricing or lease duration mismatches

3. Pipeline B: Tri-Paradigm Collaborative Filtering

3.1 Model-Based FunkSVD Matrix Factorization

The rating prediction $\hat{r}_{ui}$ for tenant u on appliance i is formulated with global, user, and item bias parameters:

$$\hat{r}(u, i) = \mu + b_u + b_i + p_u^T * q_i$$

Optimization is conducted via Stochastic Gradient Descent (SGD) with L2 regularization ($\lambda = 0.02$) over 50 epochs, achieving a test RMSE of **1.2015** across 8,924 interaction ratings.

4. Pipeline C: Local RAG with Ollama

The system retrieves matching MongoDB documents via keyword intersection, constructs an in-memory context prompt, and delegates generation to local quantized weights (e.g. `qwen2.5:0.5b` / `rentai-llm`), forcing CPU execution (`num_gpu: 0`) to prevent discrete GPU driver memory crashes.